

Subject: / /

* Propositional logic Concepts:-

↳ declarative statement

Propositional Calculus:- It is the system that deals with the method or process use for performing manipulations of the symbols according to the rules and laws

'(' ')'

① $P, Q, R \rightarrow$ atoms

② $T \quad F$
(True) (False)

③ def of logical operators

$\wedge$	AND
$\vee$	OR
$\sim$	NOT
$\rightarrow$	implies
$\leftrightarrow$	Equivalence or Biconditional

Well formed formula $\Rightarrow$ (for Propositional Cal.)

↳ atom

↳ if α W.F.F then $\sim \alpha$ also W.F.F.

Subject:

↳ if α and β is w.f.f. / /

$$\alpha \wedge \beta$$

$$\alpha \vee \beta$$

$$\alpha \rightarrow \beta$$

$$\alpha \leftrightarrow \beta$$

Q. is $\sim(\sim P \vee \sim Q)$ is it well formed formula

Yes, $P \wedge Q$

$$Q. ((\sim P \vee Q) \wedge (\sim Q) \vee P)$$

$$\hookrightarrow ((\sim P \vee Q) \wedge (\sim Q \vee P)) \quad \text{Yes.}$$

$$Q. (\sim P \vee \wedge Q) \dots \text{not w.f.f.}$$

→ Truth table Method:-

P	Q	$\sim P$	$P \wedge Q$	$P \vee Q$	$P \rightarrow Q$	$P \leftrightarrow Q$
T	T	F	T	T	T	T
T	F	F	F	T	F	F
F	T	T	F	T	T	F
F	F	T	F	F	T	T

$P \rightarrow Q$ is F if P is T & Q is F
is T otherwise

$P \leftrightarrow Q$ if both have same truth value it is T

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Precedence Rule :-

① $\sim$

② $\wedge \vee \rightarrow \leftrightarrow$

(decreasing order of priority)

$$\hookrightarrow \alpha = P \vee Q \rightarrow (\sim Q \rightarrow \sim P)$$

P	Q	$\sim Q$	$\sim P$	$P \vee Q$	$\sim Q \rightarrow \sim P$	Result (α)
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T	T	F	F	T	T	T
T	F	T	F	T	F	F
F	T	F	T	T	T	T
F	F	T	T	F	T	T

$$\hookrightarrow \underbrace{(P \rightarrow Q)}_F \wedge R \leftrightarrow \underbrace{P \wedge Q \vee S}_T \Rightarrow \textcircled{F}$$

P	Q	R	S	$P \rightarrow Q$	$\wedge R$
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F	T	F	T	T	F
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$\textcircled{F}$

$\hookrightarrow$ Show that $\sim P \vee Q$ and $P \rightarrow Q$

P	Q	$P \rightarrow Q$	$\sim P$	$\sim P \vee Q$
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T	T	T	F	T
F	T	T	T	T
T	F	F	F	F
F	F	T	T	T

$\textcircled{T}$

Subject: / /

Equivalence Laws :

① Commutative :-

$$P \wedge Q = Q \wedge P$$

$$P \vee Q = Q \vee P$$

② Association :-

$$P \wedge (Q \wedge R) = (P \wedge Q) \wedge R$$

③ $\sim \sim P \rightarrow P$

④ Distributive law :-

$$P \wedge (Q \vee R) = (P \wedge Q) \vee (P \wedge R)$$

⑤ De Morgan's law :-

$$\sim (P \wedge Q) = \sim P \vee \sim Q$$

$$\sim (P \vee Q) = \sim P \wedge \sim Q$$

⑥ Law of excluded middle

$$P \vee \sim P = T$$

⑦ Law of Contradiction

$$P \wedge \sim P = F$$

Subject:

⑧ Idempotence :-

$$P \wedge P = P$$

$$P \vee P = P$$

⑨ Absorption :

$$P \wedge (P \vee Q) = P$$

$$P \vee (P \wedge Q) = P$$

$$P \vee (\sim P \wedge Q) = P \vee Q$$

$$P \wedge (\sim P \vee Q) = P \wedge Q$$

$$P \rightarrow Q = \sim P \vee Q$$

$$P \leftrightarrow Q = (P \rightarrow Q) \wedge (Q \rightarrow P)$$

Show the following f^r are logically equivalent using the equivalence law:-

$$(i) (P \wedge Q) \vee (P \wedge \sim Q) = P$$

$$(P \wedge Q \vee P) \wedge (P \wedge Q \vee \sim Q)$$

$\downarrow$
T

$$P \wedge Q \vee \underline{P \wedge P}$$

P

$$P \wedge Q \vee P$$